IMPROVEMENT REPORT

Spare Part EMS 2.0

One platform to manage the full spare part lifecycle



Today's process loses data, time and accuracy



Fragmented data

Excel (U-dock) and the legacy system run in parallel. Only one person can edit at a time.

Missed purchase info

No bulk check of interlock, price or photos. Orders get rejected after registration.



No cost control

No real-time cost or over-budget tracking. ETA must be chased by hand.

Wrong EBOM and WO

EBOM is hard to read. Ten lines are merged into one WO on the wrong mechanism.

One platform, six connected modules

0 Forecast & Safety Stock

Predicts usage, triggers re-orders and caps stock levels.

1 Mass Search

Price, interlock, photos and 4 PO routes in one view.

2 Purchasing Flow

Multi-user requests, auto-routing and an audit trail.

3 Cost & ETA

ETA tracking, over-budget reasons, cost booked at issue.

4 EBOM

Visual line and mechanism tree with sanity rules.

5 Work Orders

Mass WO creation with correct mechanism allocation.

Before and after at a glance

Area	Before	After	Impact
0 Planning	Reactive; no forecast or safety stock	Auto forecast, reorder alerts, stock cap	**Proactive planning**
1 Search	Excel plus legacy system; no bulk lookup	Bulk lookup of price, interlock, photos, 4 PO routes	**−80% lookup time**
2 Purchasing	One editor per file; rejects get missed	Real-time multi-user; bulk handling; audit log	**No file locks, no lost orders**
3 ETA & cost	ETA via intermediaries; no live cost view	Direct ETA updates; overdue alerts; cost at issue	**Real-time budget control**
4 EBOM	Hard to read; missing groups unnoticed	Visual tree; sanity rules; fast cloning	**Standardized EBOM data**
5 Work Orders	Manual; 1 WO covers 10 lines	Bulk WO split by line and mechanism	**Accurate MTBF / MTTR**

Modules 0–1: Plan ahead, check before ordering

0 Forecast & Safety Stock

BEFORE

- No usage forecast; ordering is reactive
- Long-lead parts are ordered too late
- No safety stock: over- or under-buying

AFTER

- Auto forecast from output and time
- Reorder alerts when an order is due
- Safety-stock limit set per part
- Bulk Q-Code registration

1 Mass Search

BEFORE

- Excel and legacy system checked separately
- One item at a time; no bulk lookup
- Orders rejected after registration

AFTER

- Bulk lookup: price, interlock, stock, photos
- Status of all 4 PO routes in one view
- Item issues flagged before ordering

Module 2: Four PO routes, auto-escalated



BEFORE

- One Excel editor at a time (U-dock file lock)
- Multi-layer review: LEC, Reviewer, Requestor
- Rejects rechecked by hand; critical information gets lost
- ETA chased manually through LEC and Corporate Purchase

AFTER

- Real-time multi-user concurrent work; replaces U-dock
- LEC handles hundreds of items in bulk
- In-system comments with a complete immutable audit trail
- Overdue orders highlighted; ETA updated directly in-system

Module 3: Cost counted when parts are used

TODAY

- No real-time cost or over-budget tracking
- Cost not tied to the actual physical issue date
- Orders disconnected from stock and budget
- No view of issue-plan status per item

1 Cost on consumption

Cost is booked only when a part is issued, so each period's budget is accurate.

2 ETA & scheduling

Track arrival dates and schedule orders without cost spikes.

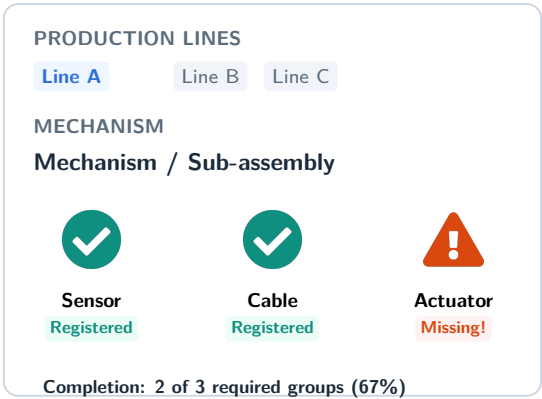
3 Over-budget control

Live budget view. A mandatory reason code is required for every overrun.

4 Issue-plan tracking

Every item shows clearly as planned, issued, or not issued.

Module 4: EBOM you can see and trust



TODAY

- No overview across parallel lines and mechanisms
- Wrong or missing part groups go completely unnoticed
- No bulk registration or visual progress tracking

WHAT CHANGES

- Visual view of all lines and mechanism trees
- Sanity rules flag/block missing groups (Sensor, Cable, Actuator)
- Mass cloning across identical lines in a single step
- Real-time completion % per mechanism and line

Module 5: One action, every work order

BEFORE

10 lines use the part

1 WO $\longrightarrow$ 1 mechanism

Line 1 shows 10$\times$ the cost; the other 9 show zero

AFTER

10 lines use the part

10 WOs, each on its own mechanism

100% precise cost allocation per line & mechanism

Less manual work

One single action automatically generates and releases all work orders simultaneously.

Correct allocation

Parts post accurately to the specific mechanism of the exact line that consumed them.

Reliable history

Issuance data matches shop floor reality for exact MTBF/MTTR and audit compliance.

BUSINESS VALUE

Impact: faster, more accurate, fully traceable

01 ▪ SPEED

60–70%

02 ▪ ACCURACY

03 ▪ TRACEABILITY

04 ▪ COST & STOCK